TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

1. Maya has \$40 and earns \$12 per hour. Which equation gives the total, y, she has after working x hours?

A. $y = 40x + 12$

B. $y = -12x + 40$

C. $\boldsymbol{y = 12x + 40}$

D. $y = 52x$

Start \$40, rate \$12 per hour

2. Jordan buys 4 notebooks that cost \$1.75 each and some pens for \$0.90 each. Write an equation for the total cost, y, when Jordan buys x pens.

$\boldsymbol{y = 0.90x + 7}$

Notebooks: $4 \cdot 1.75 = 7$; pens: $0.90x$

3. $f(b) = 3.75b + 25$. Write an inequality for sales of at most \$1150, solve it, and graph the solution on a number line.

$\boldsymbol{b \leq 300}$, closed circle at 300, shaded left

$3.75b + 25 \leq 1150 \rightarrow 3.75b \leq 1125 \rightarrow b \leq 300$